

Troubleshooting Tools

Task 1: Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

```
Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::9b69:b498:ce21:5f25%12
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2405:201:200f:c1cf:8809:a17c:6516:6cf4
    Temporary IPv6 Address. . . . . : 2405:201:200f:c1cf:7869:645b:f8a4:8f0d
    Link-local IPv6 Address . . . . . : fe80::7bb7:f9b4:f160:a70e%3
    IPv4 Address. . . . . : 192.168.29.218
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::4e93:a6ff:fe8d:56c6%3
                                192.168.29.1
```

Task 2: Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

```

C:\Windows\System32>ping www.google.com

Pinging www.google.com [2001:4860:4827:7700::] with 32 bytes of data:
Reply from 2001:4860:4827:7700::: time=26ms
Reply from 2001:4860:4827:7700::: time=21ms
Reply from 2001:4860:4827:7700::: time=25ms
Reply from 2001:4860:4827:7700::: time=22ms

Ping statistics for 2001:4860:4827:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 26ms, Average = 23ms

C:\Windows\System32>

```

Task 3: Run the `tracert` (or `traceroute` on Linux/Mac) command to `www.flipkart.com` and list the number of hops it takes to reach the destination.

```

Tracing route to flipkart.com [163.53.76.86]
over a maximum of 30 hops:

  1      2 ms      3 ms      2 ms  reliance.reliance [192.168.29.1]
  2      4 ms      4 ms      4 ms  10.7.224.1
  3      4 ms      5 ms      4 ms  172.16.12.94
  4      8 ms     10 ms      8 ms  192.168.244.244
  5      5 ms      8 ms      9 ms  192.168.232.69
  6      5 ms      8 ms      4 ms  192.168.232.98
  7      3 ms      3 ms      4 ms  192.168.244.140
  8      *          *          *    Request timed out.
  9      *          *          *    Request timed out.
 10     *          *          *    Request timed out.
 11     28 ms     29 ms     29 ms  115.247.102.134
 12     *          *          *    Request timed out.
 13     *          *          *    Request timed out.
 14     *          *          *    Request timed out.
 15     *          *          *    Request timed out.
 16     25 ms     25 ms     27 ms  163.53.76.86

Trace complete.

```

Path structure

- Hop 1 — 192.168.29.1 (reliance.reliance) — your home router/gateway.
- Hops 2–3 — 10.7.224.1, 172.16.12.94 — private IP ranges (RFC 1918), these are internal Jio/Reliance ISP infrastructure routers, likely CGNAT hops since Jio doesn't have enough public IPv4 to give every customer one.
- Hops 4–7 — 192.168.244.x, 192.168.232.x — more private-range hops, still inside the ISP's core network, routing your traffic toward their edge/peering points.

- Hops 8–10 — timed out — these routers are configured to not respond to ICMP (common for security reasons on core/backbone routers, doesn't mean the path is broken).
- Hop 11 — 115.247.102.134 — a public IP, likely a peering or exit point where traffic leaves Jio's network toward the broader internet/backbone.
- Hops 12–15 — timed out again — likely backbone or CDN-edge routers (possibly at Flipkart's hosting provider or CDN) also suppressing ICMP replies.
- Hop 16 — 163.53.76.86 — the final destination, flipkart.com's server, reached successfully.

Task 4: Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

What nslookup actually is:

A tool to query any DNS server and see how it resolves a domain name into an IP address (or other DNS records).

```
C:\Windows\System32>nslookup www.spotify.com
Server:  reliance.reliance
Address:  2405:201:200f:c1cf::c0a8:1d01

Non-authoritative answer:
Name:     atc.spotify.map.fastly.net
Addresses: 2a04:4e42:31::810
           151.101.211.42
Aliases:  www.spotify.com
```

Step by step, what actually happens:

1. When you run nslookup www.spotify.com, the PC sends a query to the configured DNS server, asking "what's the IP address for www.spotify.com?"
2. That server checks:
 - Its own cache (if it already knows the answer from a recent lookup) — this is why you got a "**non-authoritative answer**", meaning it had it cached rather than being the original source.
 - If not cached, it goes and asks other DNS servers up the chain (root servers → .com servers → the authoritative server for spotify.com) until it gets the real answer.
3. It then returns that answer back to your nslookup command, which displays it.

Task 5: A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check. Hint: Think about checking both network connectivity and DNS resolution.

- **ipconfig** — checks whether the laptop itself has a valid IP address, subnet mask, and default gateway, confirming it's properly connected to the local network at all.

- ping 8.8.8.8

Checks whether your friend's laptop has basic internet connectivity at all, by testing a known, reliable IP address (Google's DNS server) directly — bypassing DNS entirely. If this fails, the problem is with their local network/internet connection, not Zomato specifically.

- nslookup zomato.com

Checks whether the domain name is resolving to an IP address correctly. If ping 8.8.8.8 worked (so the internet connection is fine) but nslookup fails or times out, that tells you the issue is specifically a **DNS problem** — either their ISP's DNS server is down/blocking the domain, or there's a local DNS misconfiguration.

- tracert zomato.com

If DNS resolves fine (so you now have Zomato's IP), this shows the actual hop-by-hop path the traffic takes toward Zomato's server, and where it stops or slows down. This helps identify if the issue is somewhere along the route — an ISP hop timing out, a peering/backbone issue, or the final server itself not responding — rather than something on the laptop.